The Ancient Art of War — Remake

Quick Reference Guide: building, running and command-line options

The remake is written in C with SDL3 and built with CMake. It runs on macOS, and also builds on Windows and Linux. No original game files are included: the remake loads its graphics, campaigns and data from **your own copy** of the original game at run time.

What you need

Tools	CMake 3.16 or later, a C11 compiler (Clang, GCC or MSVC) and SDL3. On macOS: `brew install cmake sdl3` (Homebrew).
Game files	From the original disk: `WAR.EXE`, `M`, the `P` files, the `T` files, `TITLES`, `DTITLES`, `F`, `CNF` and `02`–`07`. Add `WARET.DAT` for the 16-colour EGA, Tandy and VGA graphics. Upper or lower case doesn't matter.

Compiling

macOS

```
cd WAR_Remake
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build
```

This makes `build/AncientArtOfWar.app`. The app icon is drawn during the build from your game files, found in `../WAR_Original_Files_3.5`. If they are elsewhere, add `-DWAR_GAME_DIR=/path/to/your/game` to the first `cmake` command. Without them the app still builds; it just has no icon in Finder. If Finder shows an old icon, run `touch build/AncientArtOfWar.app`.

Windows and Linux

The same two `cmake` commands, with SDL3 installed where CMake can find it (if it can't, add `-DCMAKE_PREFIX_PATH=/path/to/SDL3`). With Visual Studio, build with `cmake --build build --config Release`. The result is the `AncientArtOfWar` program in `build` (in `build/Release` for Visual Studio).

Tests (optional)

The test programs are in `WAR_Remake/test`. Build one with `./build.sh NAME` (with AddressSanitizer and UBSan) and run it without a window by setting `SDL_VIDEODRIVER=dummy SDL_AUDIODRIVER=dummy`. The test README lists each one and its arguments.

Running

```
open build/AncientArtOfWar.app                # macOS, from Finder or the terminal
build/AncientArtOfWar.app/Contents/MacOS/AncientArtOfWar ../WAR_Original_Files_3.5
```

On first launch the app asks for the folder that holds your game files and remembers it. It also looks in a game folder next to the app and in the current folder, and you can give the folder on the command line. Press **O** in the asset viewer to choose a different folder. Command-line options need the program inside the app bundle (`…/Contents/MacOS/AncientArtOfWar`).

Command-line options

```
AncientArtOfWar [video=CARD] [--viewer] [--log | --log=verbose] [game-folder]
AncientArtOfWar --screenshots OUTDIR [game-folder]
```

Option	What it does
`game-folder`	The folder with your original game files. Any argument that doesn't start with `-` is taken as the folder.

`video=CARD`
`--video=CARD`
`--video CARD`
	The video card to start on (plain CGA if not given). CARD is one of the names in the next table, in any case. Tab changes the card while playing.

Option	What it does
`HERC` `--herc`	Hercules monochrome, as the original's `WAR HERC`: the same as `video=herc`.
`--viewer`	Open the asset viewer (every picture, the tiles, the font and each campaign map) instead of the game. F12 on the title page opens it too.
`--log` `--log=normal`	Write a debug log of each campaign played: the enemy commander's decisions, your orders and the events.
`--log=verbose` `--log verbose`	The same, with every candidate the enemy weighed, its cell tables and a table of all squads every 16 ticks.
`--screenshots OUTDIR`	Write every screen as BMP files for each card into OUTDIR, then quit (no window).

Video cards (video=CARD)

CARD	Card	What you see
`cga`	CGA (default)	WAR.EXE's 320×200 in 4 colours: cyan, magenta and white; the scrolls in yellow, red and green
`composite`, `cga_composite`	CGA composite	The CGA picture as the original IBM CGA shows it on a TV or composite monitor: 640 dots across, colours made by each pixel's neighbours
`composite2`, `cga_composite2`	CGA composite (new)	The same on IBM's later CGA, whose colours are paler
`ega`	EGA	WARET.EXE's 16-colour pictures in the standard 16 colours *
`tandy`	Tandy 1000	The standard 16 colours, with the Tandy's own pointer, yellow cursor and outlines *
`vga`	VGA	Each picture in its own palette, 6 bits a channel, as the VGA showed it *
`herc`, `hercules`	Hercules	The CGA picture in monochrome, each pixel as two dots, in the middle of a 720×348 screen

* Needs `WARET.DAT`. Without it, Tab skips these cards and `video=` falls back to CGA. Tab goes through the cards in the order of this table.

> **Options Shoppe and the CGA cards:** on a CGA card, the Options Shoppe's display line *tv/composite* switches to CGA composite and *rgb* to the plain CGA; *black/white* turns the picture grey. On EGA, Tandy and VGA the line just names the card, as in the original 16-colour program.

Examples

```
APP=build/AncientArtOfWar.app/Contents/MacOS/AncientArtOfWar   # macOS
$APP video=ega                        # start on an EGA
$APP video=composite                  # the CGA on a TV
$APP HERC                             # as WAR HERC
$APP --log=verbose ~/Games/WAR        # a detailed log; game files in ~/Games/WAR
$APP --viewer                         # the asset viewer
$APP --screenshots ~/Desktop/shots ../WAR_Original_Files_3.5
```

Keys that work everywhere

Key	Action
Tab	Next video card
F2	The original monitor’s 4:3 shape, or square pixels
⌘Enter on a Mac; Alt+Enter on Windows and Linux	Full screen
F12 (title page)	The asset viewer
Esc	Back up in the menus; during a game, leave it (Y or N)
⌥S on a Mac (Option+S); Alt+S on Windows and Linux, during a game	Save the game to the data disk

All the original game keys work as they did in the original; the remake’s README lists them in full.

Where the remake keeps its files

macOS: `~/Library/Application Support/AncientArtOfWar/Remake/` (on Windows and Linux, SDL’s per-user data folder for the app).

File or folder	Contents
`CNF`	The Options Shoppe’s settings, in the original’s 25-byte layout
`DRIVE`	Which data disk is “in the drive”
`F`	Formations you have taught (once you save one)
`Data Disks/<name>/`	One folder per data disk (up to 12): TITLES, the T files of its campaigns, and its saved game S. A game saved by the original (S, or SET from the 16-colour program) can be carried on: copy the floppy’s files into a new folder here.
`Logs/`	The `--log` files, one per campaign, named by date, time and campaign